MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5814A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, medium Mu

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Normal:

Parameter:	Ef	Eb	Ec	Ehk	Rg/g	Ik/k	Ic/g	Pp/p	TE	Alt
Unit:	V	Vdc	Vdc	v	Meg	mAdc	mAdc	W	°C	ft
Maximum:	13.8	330	0,-55	100	0.5	22	5.0	3.0	165	Note 1
	6.9	---	---	---	Note 2	---	---	---	---	---
Minimum:	11.4	---	---	---	---	---	---	---	---	---
	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS (1):	12.6	250	-8.5	0	---	---	---	---	---	---

Pulse service:

Parameter:	Ef	Eb	Ec	eb	egk	ehk	Pg/g	Pp/p	ik/k	Du	tp	prr	Rk/k
Unit:	V	Vdc	Vdc	v	v	v	W	W	ma	%	µs	---	Ohms
Maximum:	13.8	330	-55	660	+60	100	0.28	3.0					---
	6.9	---	---	---	-100	---	---	---					---
Minimum:	11.4	---	---	---	---	---	---	---	See figure 1				---
	5.7	---	---	---	---	---	---	---					---
TEST CONDITIONS (2):	12.6	250	-45	---	+55	0	---	---	---	1.0	10	1,000	1.0

GENERAL:

Qualification

Reliable tube

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	160	190	mA
1336	Heater-cathode leakage	Test condition (1) (see note 3)	0.4	II	Ihk	---	7	µAdc
1266	Total grid current	Test condition (1); Rg/g = 0.5 Meg (see notes 3 and 4)	0.4	II	Ic	0	-0.5	µAdc
1256	Electrode current (1) (anode)	Test condition (1) (see note 3)	0.4	II	Ib	6.5	14.5	mAdc
1256	Electrode current (2) (anode)	Test condition (1); Ec = -30 Vdc; Rp = 0.1 Meg (see note 5)	0.4	II	Ib	---	20	µAdc
1306	Transconductance (1)	See note 3	0.4	II	Sm	1,750	2,650	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 3	2.5	I	R	1,000	---	Meg
1256	Electrode current (1) (anode) (difference between sections)		2.5	I	Ib	---	3.5	mAdc
1256	Electrode current (3) (anode)	Ec = -18 Vdc (see note 3)	2.5	I	Ib	5	---	µAdc
1306	Transconductance (2)	Ef = 11.4 V (see note 3)	2.5	I	ΔSm Ef	---	10	%
1266	Grid emission	Ef = 15 V; Ec = -30 Vdc; Rg/g = 0.5 Meg (see notes 3 and 6)	2.5	I	Isg	0	-1.5	µAdc
2201	Noise and microphonics	Ef = 12.6 V; Ebb = 300 Vdc; Ec = 0; Ecal = 50 mVac; Rk = 1,500 ohms; Rp = 50,000 ohms; Rg = 0 (see note 7)	2.5	I	---	---	---	---
1316	Amplification factor	See note 3	2.5	I	Mu	15.5	18.5	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1296	Pulse current (1) (cathode)	Test condition (2) (see notes 3 and 8)	2.5	Code H	ik Δik (tp)	400 ---	--- 15	mA %
1296	Pulse current (2) (cathode)	Test condition (2); Ef = 11.4 V (see notes 3 and 8)	6.5	Code H	ik Δik (tp)	350 ---	--- 20	mA %
1306	Transconductance (3)	Eb = 100 Vdc; Ec = 0 (see note 3)	6.5	Code H	Sm	2500	4000	µmhos
1331	Direct-interelectrode capacitance	No shield (note 3) No shield (note 3) No shield, sec 1 No shield, sec 2	6.5	Code E	Cgp Cin Cout Cout	1.20 1.25 0.30 0.20	1.80 1.95 0.70 0.60	pF pF pF pF
1031	High-frequency vibration	Rp = 2,000 ohms; (see note 7)	6.5	Code H	Ep	---	100	mVac
1031	Sweep-frequency vibration	Rp = 2,000 ohms (see note 7)	6.5	Note 9	Ep	---	300	mVac
1041	Shock	450 G; Ehk = +100 Vdc (see note 10)	6.5	Note 9	---	---	---	---
1031	Vibration fatigue		6.5	Note 9	---	---	---	---
---	Post-shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	150	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	15	µAdc
1306	Transconductance (1)		---	---	Sm	1,600	---	µmhos
1266	Total grid current		---	---	Ic	0	-1.5	µAdc
1121	Base strain	See note 11	---	---	---	---	---	---
2126	Envelope strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V (heaters in parallel); Ehk = +135 Vdc; Ec = Eb = 0; 1 min "on", 4 min "off"	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	15	µAdc
1516	Stability life	Test condition (1); Ehk = 135 Vdc; Rg/g = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end points:					
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	10	%
1201	Short and discontinuity detection		---	---	---	---
1306	Transconductance (1)		Sm	1,600	---	µmhos
1501	Intermittent life	Test condition (1); Ehk = 135 Vdc; Rg/g = 0.5 Meg; TA = room; TE = +165°C (min) (see note 12); Group E	---	---	---	---
---	Intermittent life-test end points (1,000 hours):					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	0	-0.5	µAdc
1301	Heater current		If	160	196	mA
1306	Change in transconductance (1) of individual tubes		ΔSm t	---	25	%
1306	Transconductance (2)		ΔSm Ef	---	20	%
1336	Heater-cathode leakage		Ihk	---	7	µAdc
1211	Insulation of electrodes		R	250	---	Meg
1501	Intermittent life (pulse)	Test condition (2); Eb = 300 Vdc; Ec = -40 Vdc; egk = +60 ±3 volts; Rb/p = 180 ohms; Rg/g = 47 ohms (see note 13); Group E	---	---	---	---
---	Intermittent pulse life-test end points (1,000 hours):					
---	Inoperatives		---	---	---	---
1296	Pulse current (1) (cathode)		ik	250	---	ma

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 3 -Continued					
1511	Cathode interface life	Ef = 6.9 V; heaters in parallel; (see note 3)	Ri	---	25	Ohms

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This value is for operation under fixed bias conditions. With cathode bias, Rg may be 1.0 megohm maximum.
3. Test each unit separately.
4. This test shall be performed at the conclusion of the holding period.
5. Test each unit separately with normal test condition voltages applied to the opposite section.
6. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Eb	Rk/k	Rg/g
V	Vdc	Vdc	Ohms	Meg
15.0	-8.5	250	0	0.5

7. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
8. The grid pulse shall be a square wave meeting pulse shape requirement of method 1296 and in addition, the maximum amplitude shall occur within the first 20 percent of tp; tp = 10 µs and prr = 1,000 pps. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to the positive end of the cathode resistor. Grid resistance, not exceeding 50 ohms, may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal.

   Peak currents shall be measured by means of a high impedance oscilloscope, or equivalent device, connected across a cathode resistor of 1.0 ±0.1 ohm. The specified limit refers to the maximum of the pulse amplitude. The variation of the output pulse amplitude between 20 percent tp and 80 percent tp shall not exceed the specified limits for $\Delta ik_{(tp)}$. Preheat at specified Ef for 5 minutes; no other voltages applied. (See figure 1.)
9. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
10. A grid resistor of 0.1 megohm shall be added to each section; however, this resistor shall not be used when a thyratron-type short indicator is employed.

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11. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature, sampling (method 1121)", except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

    (a) Accepted if not more than one defective for class "A", "B", or "C" defects, respectively, (see method 1121), or if not more than a total of two defectives are found in the sample.

    (b) Rejected if two or more defectives for class "A", "B", or "C" defects, respectively, or if a total of three or more defectives are found in the sample.

12. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

13. The positive portion of the grid pulse shall be a rectangular wave meeting pulse shape requirement of method 1296. The pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminal with respect to the cathode. Grid resistance not exceeding 50 ohms may be inserted to prevent oscillation, provided readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. The pulse width, tp, shall be 10 ±2 µs, and the duty factor, 0.9 percent to 1.1 percent. Self-excited life-test circuitry is permissible, provided any additional anode voltage drops during the time of the pulse are compensated for by increasing Ebb. No fixed Ecl need be applied under self-excited conditions.

14. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the change.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR
Air Force - 99
DLA - ES

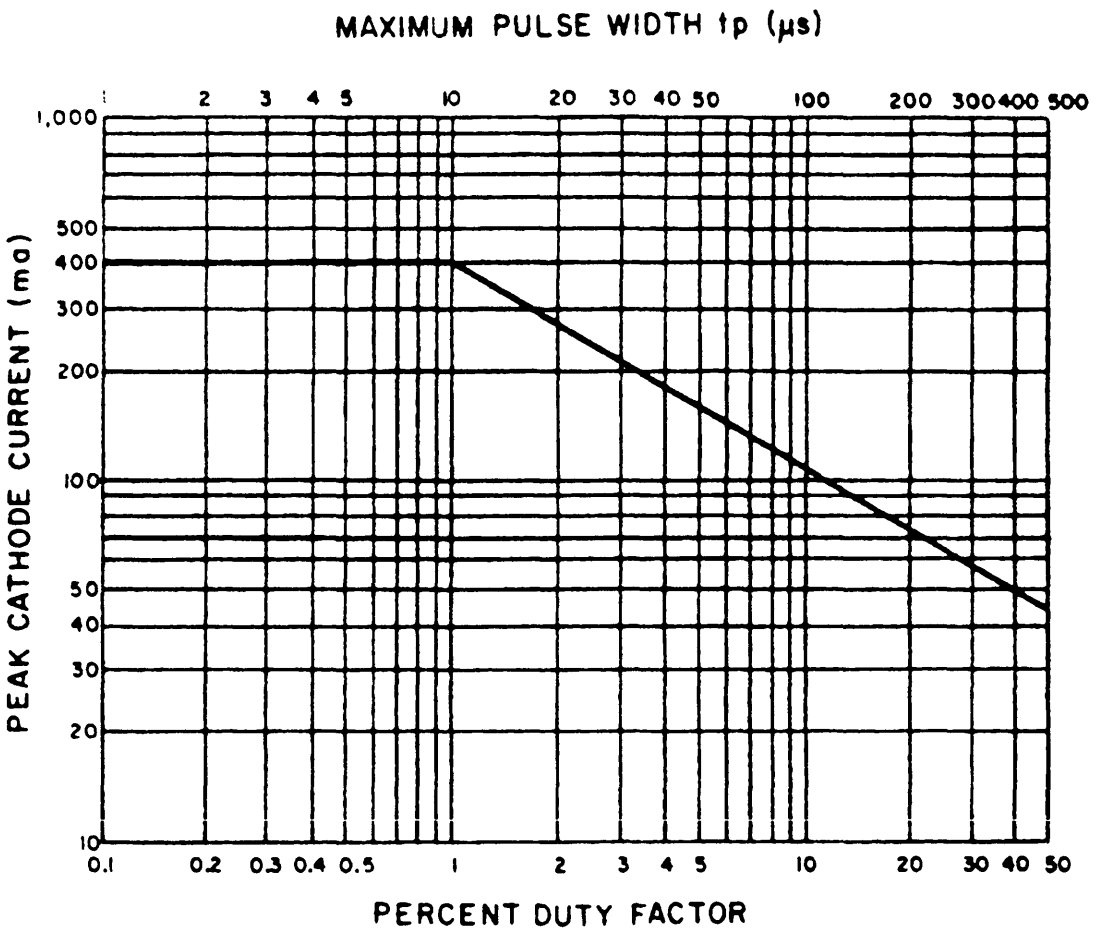
User activities:
Army - AV
Navy - SH, AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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The area below and to the left of the line is the area of permissible operation. Four hundred milliamperes is the maximum peak current at any duty factor less than 0.1 percent. For any tube operating under the worst probable conditions, the point indicating peak cathode current and percent duty factor and the point indicating peak cathode current and pulse width should both be in the area of permissible operation. Duty factor is defined as the ratio of the average current to the maximum peak current occurring in any 1,000-microsecond period.

Figure 3. Pulse rating chart.

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